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Location of the western Pontides, NW Turkey, during Triassic time: preliminary palaeomagnetic results

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SUMMARY

This paper reports preliminary results of palaeomagnetic data obtained from the lower-Triassic (Schtian) and mid-Triassic (Anisian) rocks in the western Pontides, NW Turkey. The Triassic rocks have been studied by applying thermal and/or alternating-field demagnetization. The characteristic magnetization has a 50° mean inclination derived from 41 samples taken from three separate sites. Although the inclination values for the three sites are similar, the corresponding declination angles indicate that the Gümüşdere site, which lies to the north of Istanbul city, has undergone 68° anticlockwise rotation; while the Tavşancıl and Denizliköy sites, which are to the southeast of Istanbul, have rotated 40° clockwise on average. The mean inclination (50°) places the western Pontides at the southern margin of Laurasia during lower- and mid-Triassic time.

Key words: palaeomagnetism, Triassic, Turkey, western Pontides, Laurasia

1 INTRODUCTION

The determination of the palaeolatitude of the Pontides in northern Turkey is of critical importance in assessing whether the Pontides belonged to Gondwana or Laurasia during early- to mid-Triassic time. The early palaeomagnetic work of the Amasra redbeds of Permo-Triassic age by Gregor & Zijderveld (1964) in the northwestern Pontides indicated a magnetization direction of $D = 292°$ and $I = -14.8°$. Lauer (1981) pointed out that this magnetization might have been acquired south of the equator so that the Pontides could be interpreted as parts of Gondwana during Permo-Triassic time. However, the Permo-Triassic inclination data can be interpreted as of either normal or reversed polarity as Lauer (1981) also noted. Thus, a southerly origin of the Pontides cannot be confirmed solely from this palaeomagnetic work. From the geological point of view, Şengör (1979), Şengör & Yılmaz (1981) and Şengör, Yılmaz & Sungurlu (1985) argued that the Rhodope–Pontide fragment was part of the Gondwana continent until the mid-Triassic. However, Şengör (1987) later suggested that the Istanbul–Balkan fragment may have been transported from southern Laurasia to northern Gondwana in the early-Permian–mid-Triassic interval. According to Robertson & Dixon (1985), the Rhodope–Pontide fragment belonged to Laurasia rather than Gondwana during all geological times. Another model, in which Turkey, south of the Izmir–Ankara–Erzincan suture remained part of Gondwana until late Cretaceous time, has been developed by Adamia *et al.* (1980). To clarify this controversy, we sampled all the known exposures of early Triassic volcanic rocks in the western Pontides, NW Turkey, for palaeomagnetic measurements (Fig. 1). The results of this palaeomagnetic investigation provide evidence that at least the western Pontides (Istanbul Nappe: Şengör *et al.* 1985) was part of Laurasia during early- to mid-Triassic time.

2 GEOLOGY OF THE TRIASSIC SITES

We sampled three sites of Triassic rocks. At the first site, which is north of Istanbul near the Kilyos area (Fig. 1), the sampled basaltic flow of Schtian age underlies lower Carboniferous rocks in an overturned limb (Kaya & Lys 1982). At the second site, which is located in the Gebze area near Tavşancıl village SE of Istanbul, the sampled andesitic unit is interbedded with redbeds of Schtian age (Özdemir, Talay & Yurtsever 1973, Fig. 1) which rests with a sharp unconformity on moderately to strongly folded Devonian and Carboniferous rocks. At the third site, which is also located in the Gebze area 20 km N of Tavşancıl near Denizliköy village, the sampled unit is limestone of Anisian age overlying redbeds of Schtian age (Aytuna 1974, see also Fig. 1).

3 SAMPLING AND MEASUREMENT PROCEDURES

Forty-one samples were collected from three sites by hammer/chisel and by drilling methods. Samples were oriented by both magnetic and sun compasses. The

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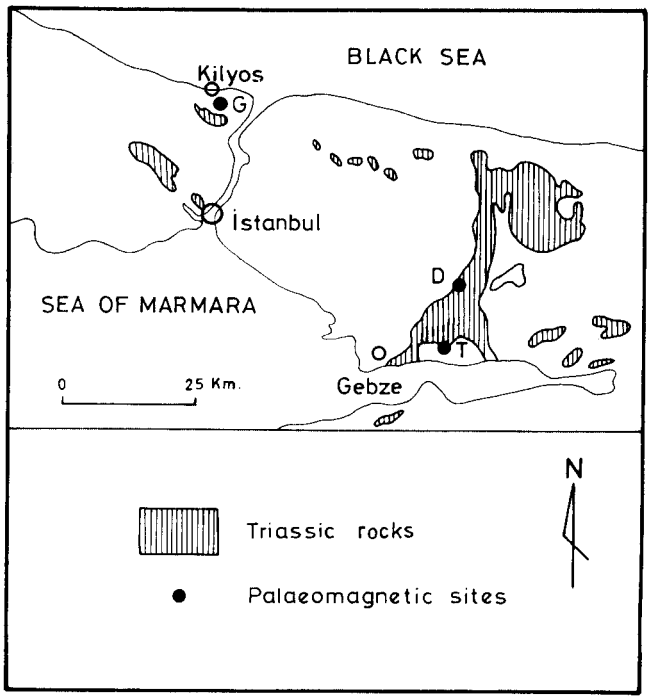


Figure 1. Location of palaeomagnetic sites in NW Pontides in Turkey. T. Tavşancıl; D. Denizliköy; G. Gümüşdere. Distribution of total Triassic rocks from Kaya & Lys (1982).

palaeomagnetic measurements were carried out at the Kantek laboratory, which is run by the İstanbul Technical University and Kandilli Observatory, and Centre des Faibles Radioactivités, Gift-sur-Yvette in France. In order to analyse the natural remanent magnetization, all samples were progressively demagnetized by thermal and/or alternating-field demagnetization procedures. The results were analysed using orthogonal and equal-area nets and normalized intensity curves.

4 PALAEOMAGNETIC ANALYSIS

4.1 The Gümüşdere site (GÜ01)

Fourteen samples were taken from Gümüşdere village near the Kilyos area. To determine their stable remanent magnetizations, samples were demagnetized in 50 °C temperature steps up to 600 °C (Fig. 2). All samples lost their secondary magnetizations below 300 °C; a higher temperature component remained stable to 600 °C. The intensities of these samples also slowly decreased after 300 °C, without any change in the direction of magnetization. Alternating-field demagnetization was not so successful as heating in eliminating secondary components.

After having applied thermal cleaning and a tectonic correction, the mean magnetization direction becomes $N = 14$, $D = 292.1°$, $I = +47.1°$, with $k = 187.0$ and $\alpha_{95} = 2.9°$ (Fig. 3, see also Table 1).

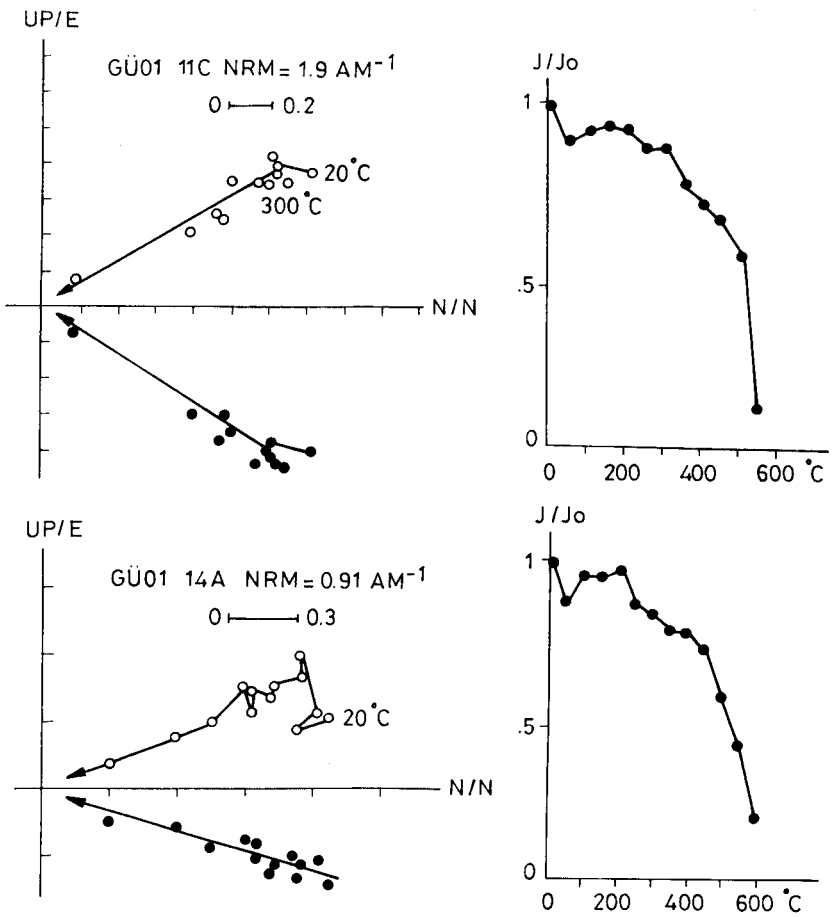


Figure 2. Typical thermal demagnetization diagrams from the Gümüşdere site. Full circles: horizontal projection; open circles: vertical projection. J_0: NRM intensity.

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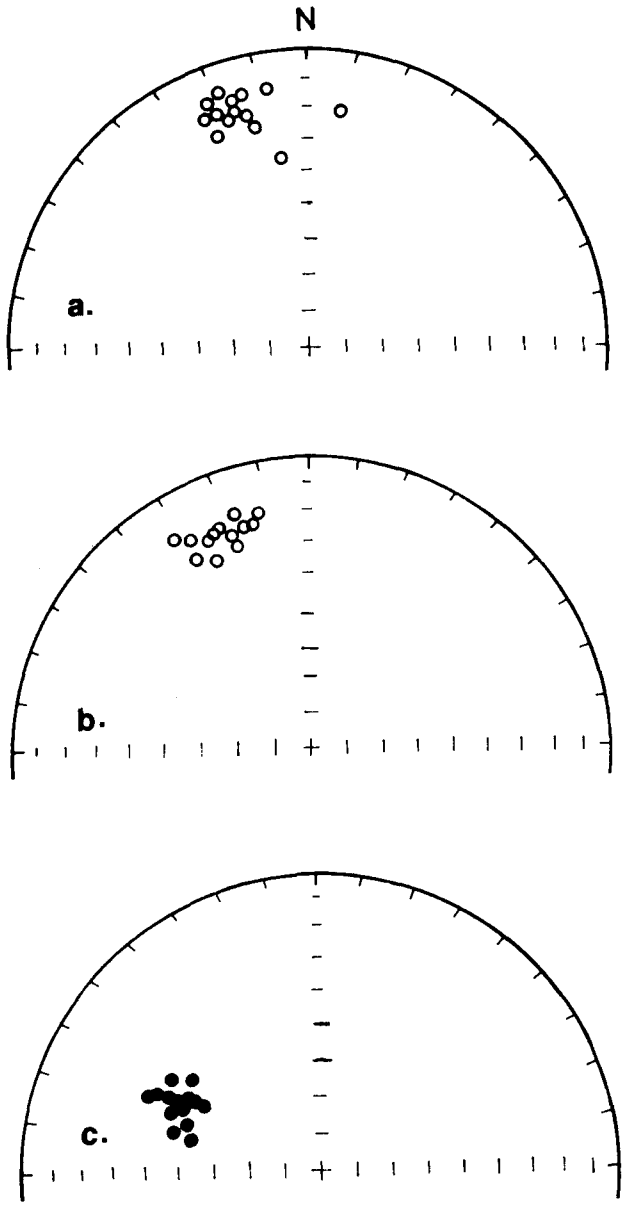


Figure 3. Stereographic projections of the directions of magnetization from the Gümüşdere site. (a) NRM; (b) after demagnetization; (c) after demagnetization and tectonic correction. Full circles: normal directions; open circles: reverse directions.

4.2 The Tavşancıl site (TA02)

Seventeen samples were collected from an andesitic flow interbedded with redbeds. Alternating-field demagnetization failed to remove supposedly existing secondary magnetizations, whereas such secondary magnetizations were eliminated with thermal cleaning (Fig. 4). In general, results of thermal cleaning indicate the presence of two components, one of which was removed at about 200–300 °C; the other component, however, remained stable between 200–700 °C. The mineral responsible for the latter magnetization component is thus inferred to be hematite. In addition, three samples, which were obtained by drilling, indicate a different magnetization history: thermal cleaning showed that they possess two components with non-overlapping coercivity spectra. The relatively softer magnetization is destroyed at about 500 °C, above which the harder magnetization direction exhibits itself between 500 and 700 °C (Fig. 4). After thermal cleaning of all samples and tectonic correction, the mean magnetization direction becomes $N = 17$, $D = 228.9°$, $I = -55.2°$, with $k = 49.0$ and $\alpha_{95} = 5.1°$ (Fig. 5, see Table 1).

4.3 The Denizliköy site (DE01)

Ten samples were analysed from the limestone of Anisian age at this site. Stepwise thermal demagnetization was used for separating secondary and primary directions. The secondary component was cleaned at 200 °C and below, while the primary component persisted without any direction between 200 and 450 °C (Fig. 6). After thermal cleaning and tectonic correction, the mean magnetization direction of samples becomes $N = 10$, $D = 34.5°$, $I = +47.4°$ with $k = 117.9$ and $\alpha_{95} = 4.5°$ (Fig. 7, see Table 1).

5 PALAEOMAGNETIC AND GEOLOGICAL CONCLUSIONS

The Triassic declination and inclination angles obtained from the sites are shown in Fig. 8. The similarity in inclination at all three sites is quite good. The declination angles differ in two ways: the Gümüşdere site has rotated 68° anticlockwise, whereas the Tavşancıl and Denizliköy sites have rotated 40° clockwise on average. At the Tavşancıl and Denizliköy sites the normal and reversed directions are antipodal, indicating good magnetic stability

Table 1. Palaeomagnetic results from the Triassic rocks. (**a**) NRM and RM after thermal demagnetization; (**b**) RM after tectonic correction

a

Site	Age	n/N	D	I	k	α_{95}	D	I	k	α_{95}
GÜ01	Schtian	14/16	345.4	−16.8	68.8	4.8	339.7	−22.3	186.6	2.9
TA02	Schtian	17/17	262.3	−42.4	3.4	22.7	264.1	−38.3	49.0	5.1
DE01	Anisian	10/10	30.5	+62.2	30.1	9.0	42.0	+66.8	118.0	4.1

b

Site	Strike and dip	After tectonic correction			
		D	I	k	α_{95}
GÜO1	297, 56 SW (upside-down)	292.1	+47.1	187.0	2.9
TA02	220, 35 NW	228.9	−55.2	49.0	5.1
DE01	295, 20 NE	34.5	+47.4	117.9	4.5

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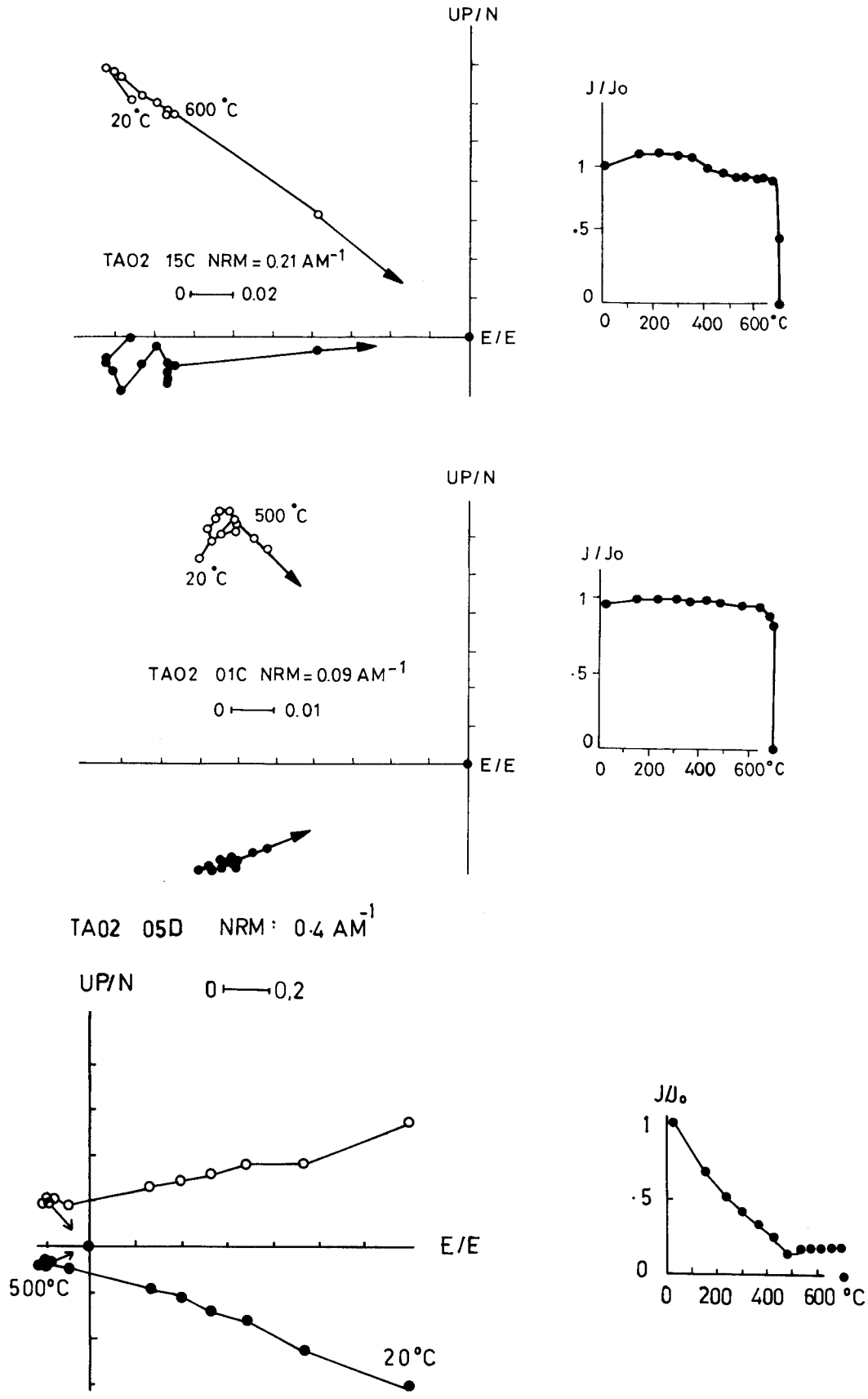


Figure 4. Typical thermal demagnetization diagrams from the Tavşancıl site.

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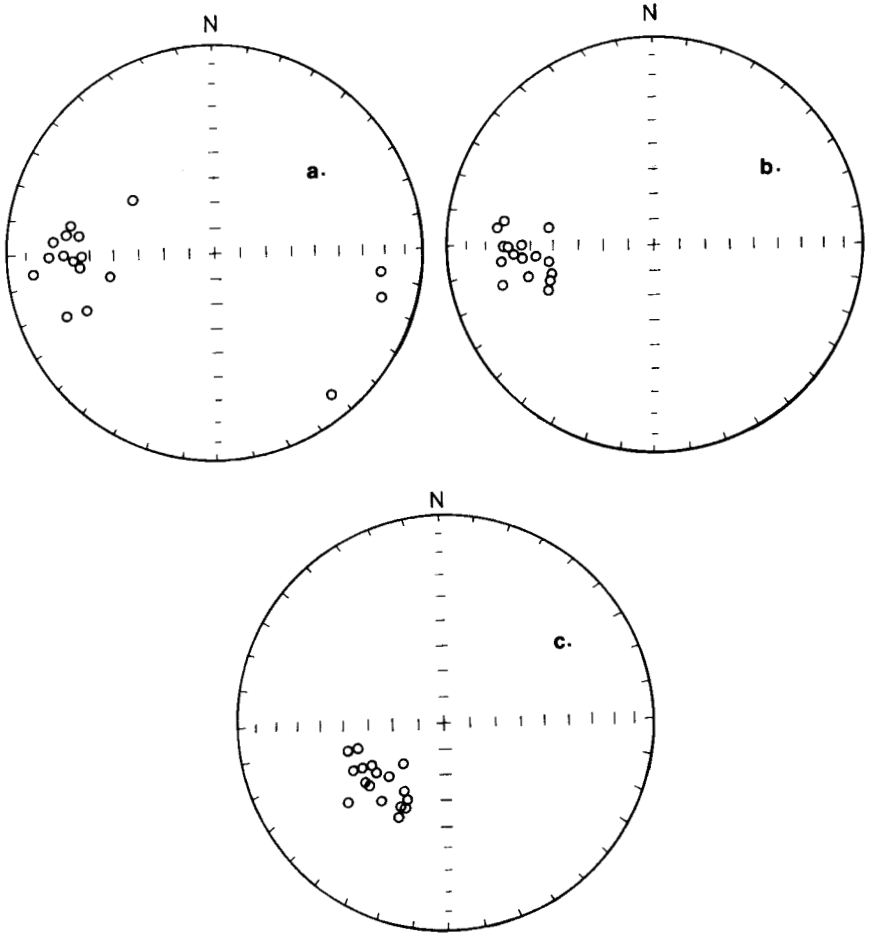


Figure 5. Stereographic projections of results from the Tavşancıl site. For explanation, see Fig. 3.

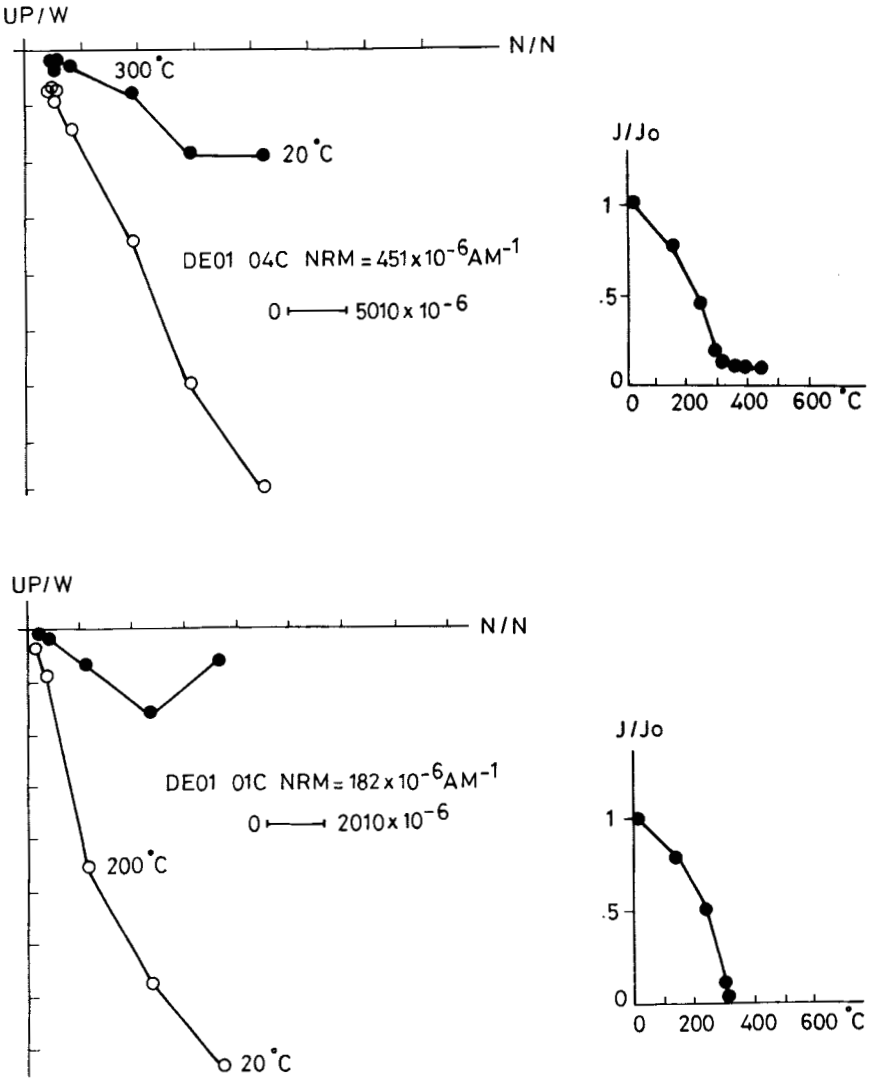


Figure 6. Typical thermal demagnetization diagrams from the Denizliköy site.

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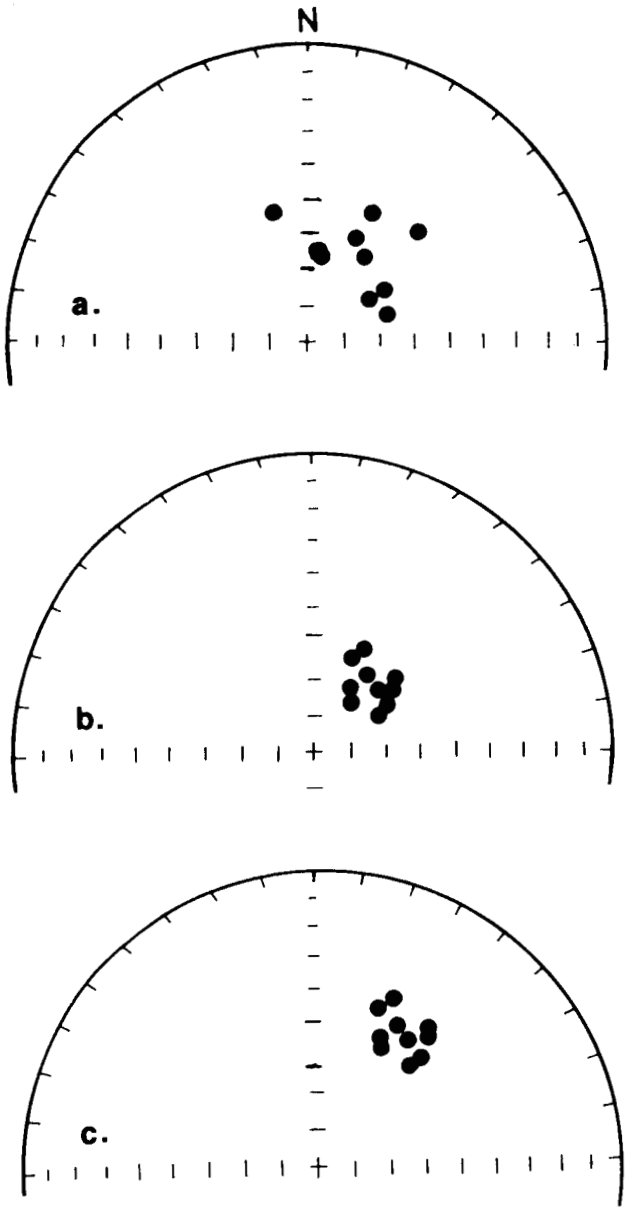


Figure 7. Stereographic projections of results from the Denizliköy site. For explanation see Fig. 3.

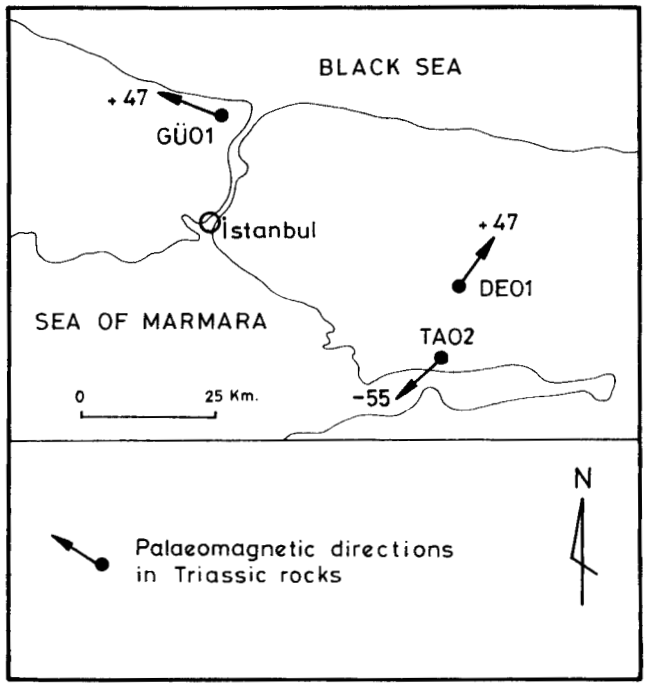


Figure 8. Triassic palaeodeclinations and inclinations at the sites in western Pontides.

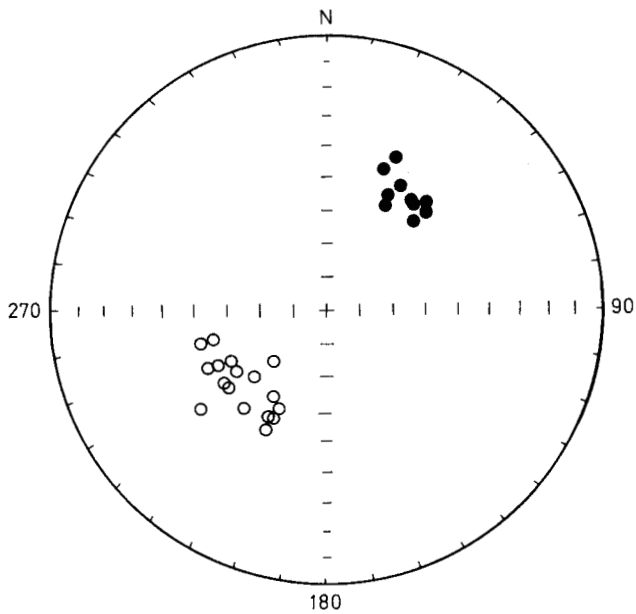


Figure 9. Comparison of the magnetization directions of the Tavşancıl and Denizliköy sites. The opposed normal and reverse directions are proof of the magnetic stability and primary nature of the remanent magnetization.

and satisfactory cleaning (Fig. 9). The Gümüşdere sites does not fit the other two sites, possibly due to some sort of flake rotation.

The palaeomagnetic results suggest that the western Pontides were at 30° palaeolatitude during early to mid-Triassic time. The 30° palaeolatitude fits better with the Eurasian palaeolatitude curve than with the African palaeolatitude curve (Westphal *et al.* 1986, Fig. 10). The palaeomagnetic work on the Amasra redbeds of Permo-Triassic age by Gregor & Zijderveld (1964) in northwestern Turkey has been the subject of some debate as to whether the magnetization was acquired south or north of the equator (Lauer 1981, Robertson & Dixon 1985), an important argument because the Pontides would then be interpreted either as having been part of Gondwana or Laurasia (respectively). Taking into account the palaeomagnetic result of this work and that of Gregor & Zijderveld and since the Amasra region is also located on the western Pontides, it can be said that the western Pontides belonged to Laurasia rather than Gondwana between Permian and mid-Triassic time. With this interpretation, the inclination of Permo-Triassic data is assumed to have been acquired

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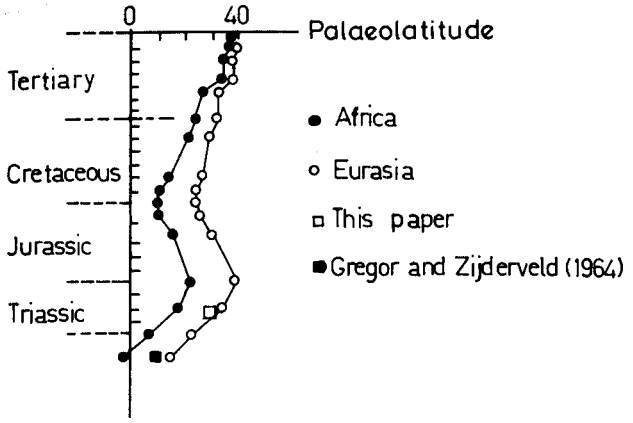


Figure 10. Comparison of the Triassic palaeolatitude of the western Pontides with the African and Eurasian palaeolatitude curves. The curves are taken after Westphal *et al.* (1986).

north of the equator. All the early-Mesozoic magnetic directions on the western Pontides are shown in Fig. 11.

The palaeotectonic evolution of the western Pontides can be elaborated in two geological time divisions on the basis of known palaeomagnetic and geological data.

(i) the western Pontides were part of Laurasia and were at 7° palaeolatitude during Permo-Triassic time.

(ii) the western Pontides then moved up with Laurasia to 30° palaeolatitude between Permian and mid-Triassic time. The existence of the Hercynian deformation in the western Pontides and the discovery of plants similar to those of Laurasia (Kerey 1984) in the western Pontides strongly support our palaeomagnetic conclusion.

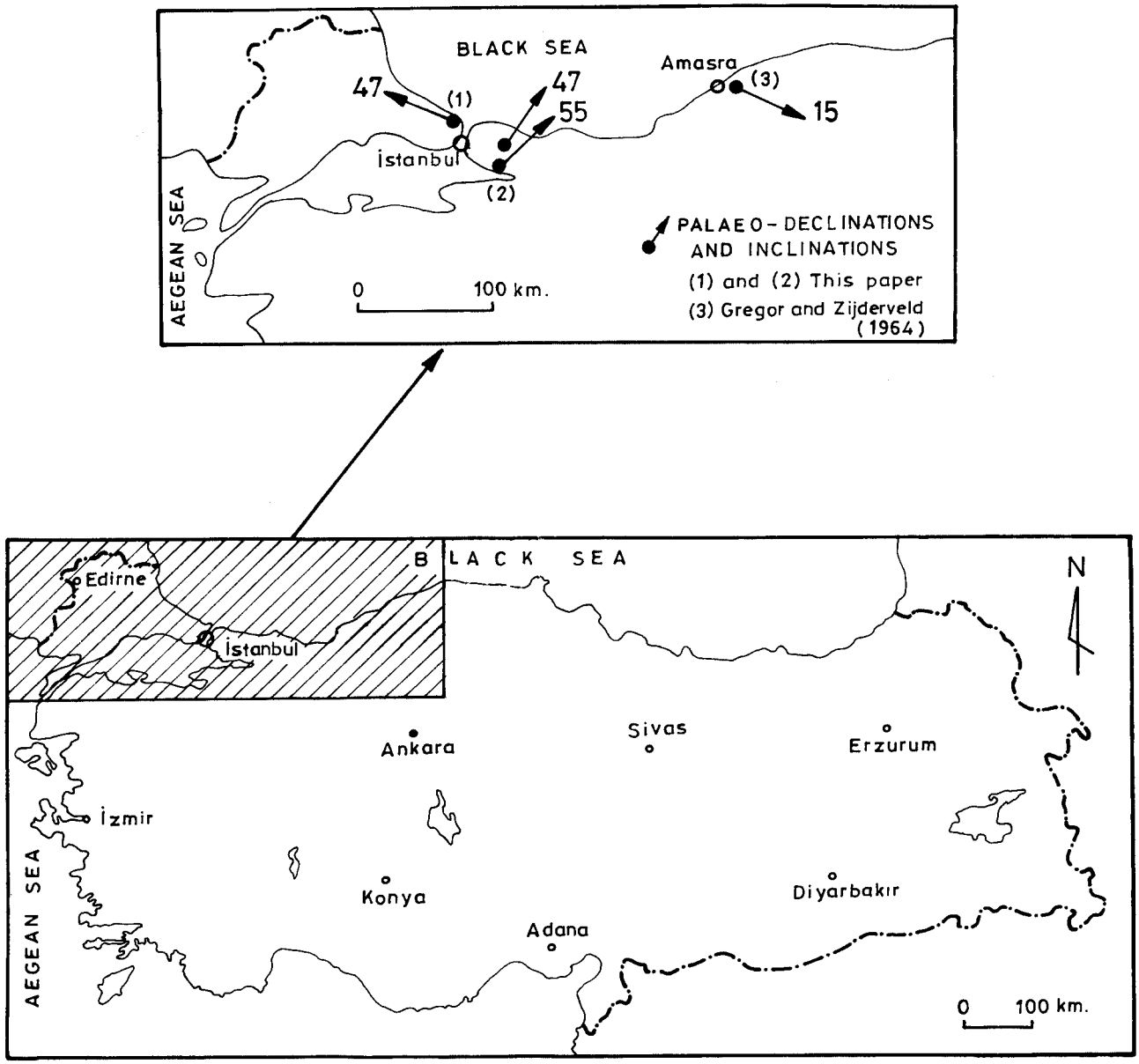


Figure 11. All the early Mesozoic palaeomagnetic data in the western Pontides. The reverse directions have been inverted through the origin.

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